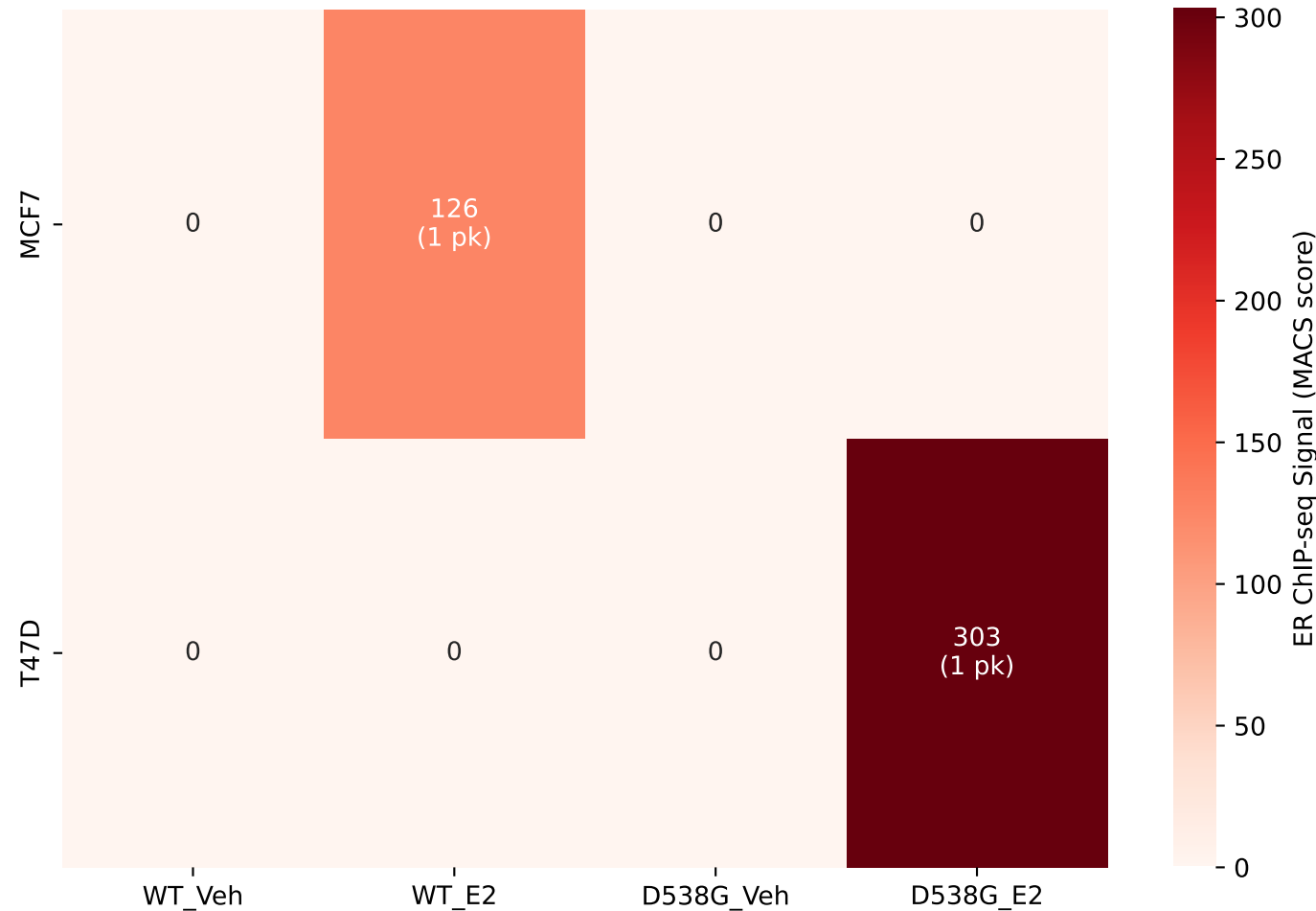
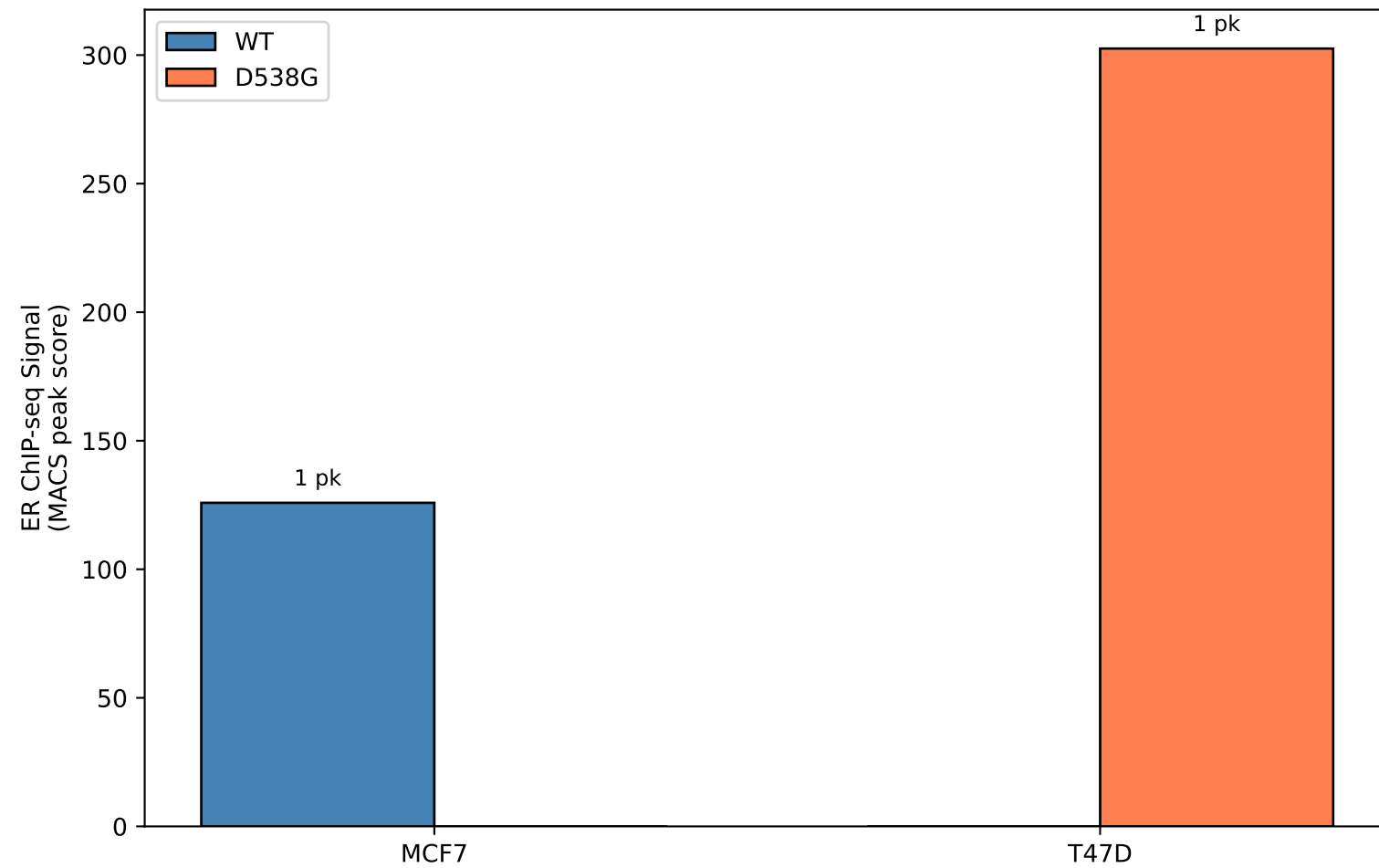


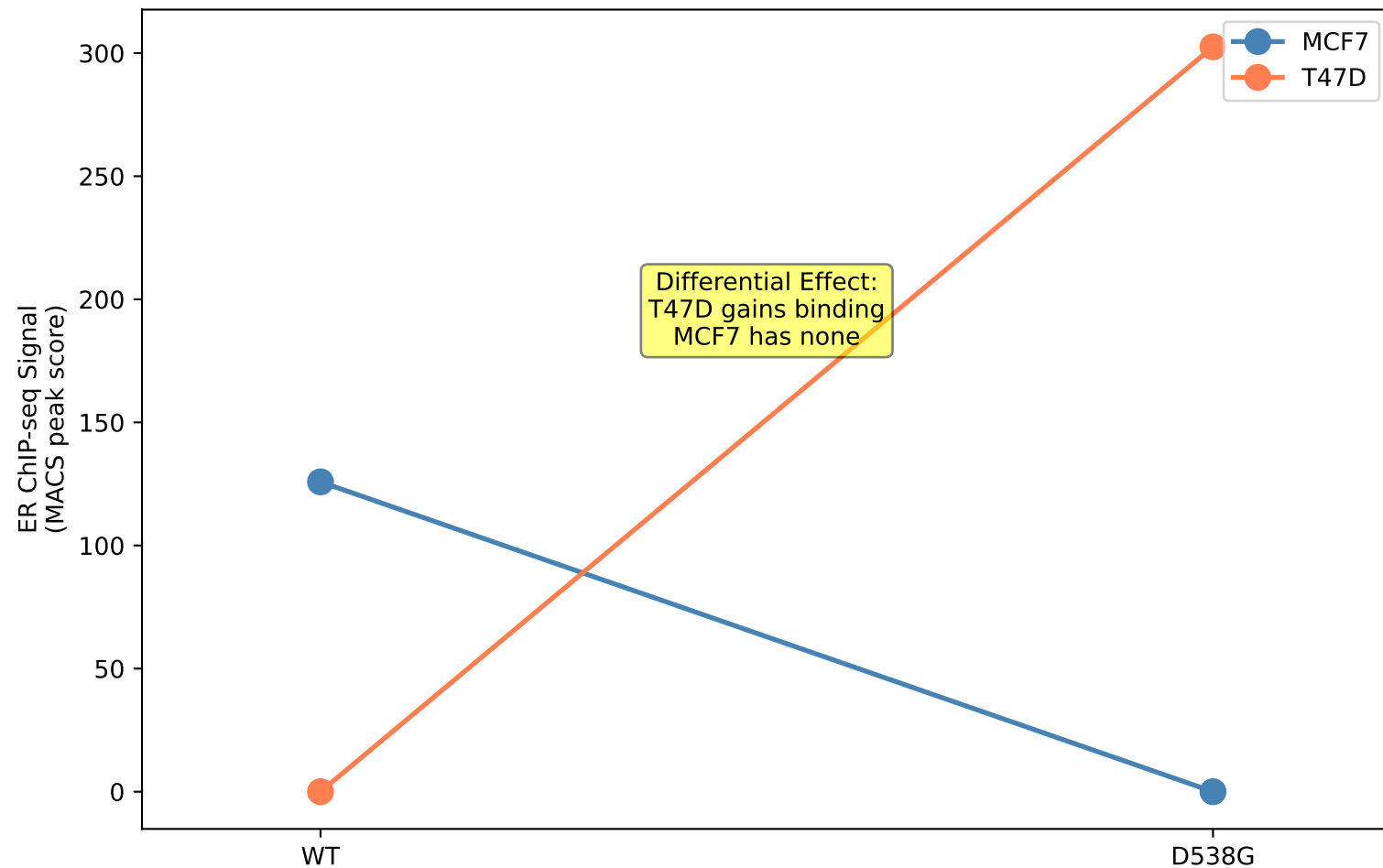
A. ER ChIP-seq Signal at HSP90B1
(GSE125117)



B. D538G Effect on ER Binding at HSP90B1
(GSE125117, E2 condition)



C. Cell Line × Genotype Interaction
(GSE125117)



DIFFERENTIAL ER BINDING AT HSP90B1 (GSE125117)

E2 CONDITION - ChIP-seq Signal (MACS score):

MCF7:

WT: 1 peak (signal=126)
D538G: 0 peaks (signal=0)
→ NO CHANGE in binding

T47D:

WT: 0 peaks (signal=0)
D538G: 1 peak (signal=303)
→ D538G GAINS binding

INTERPRETATION:

- T47D: open chromatin at HSP90B1 (ATAC-seq 3.1x)
→ ER-D538G binds → REPRESSES HSP90B1
- MCF7: closed chromatin at HSP90B1
→ ER-D538G cannot bind → NO repression

This explains HSP90B1 expression:

- T47D-D538G: DOWN (ER represses it)
- MCF7-D538G: UP (escapes ER repression)